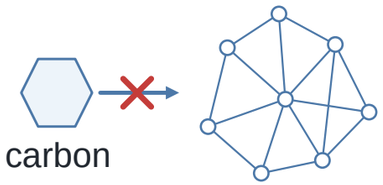
a

No-carbon phenotype screen



Carbon-source uptake closed

Objective screening

Positive objectives

Reducing equivalents

○ **NADH** **665.802**

○ **NTPs** **96.250 each**

ATP / CTP / GTP
UTP / ITP

○ **Urea** **5.000**

○ **Allophanate** **5.000**

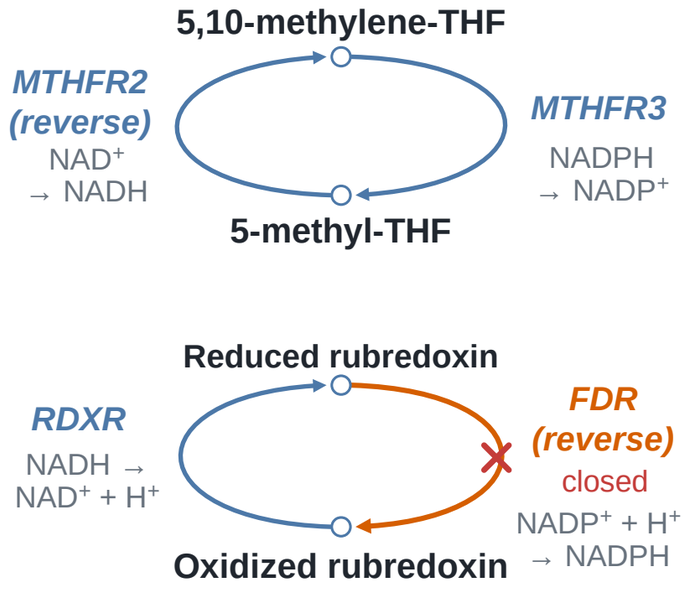
Trace active pathways

mmol gDCW^{-1} h^{-1}

b

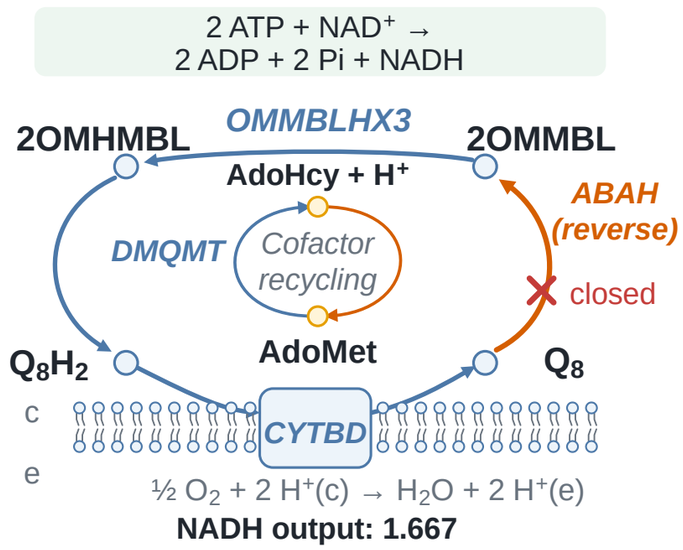
Active pathways and targeted constraints

(1) FDR-dependent redox cycle



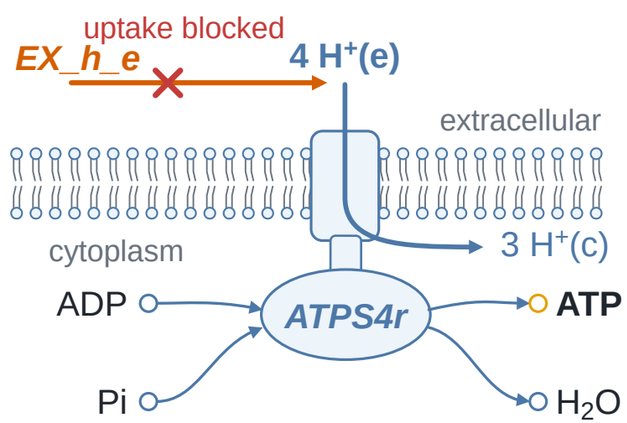
Net: $H^+ + NAD^+ \rightarrow NADH$; output 500

(2) ABAH-dependent redox cycle



NADH output: 1.667

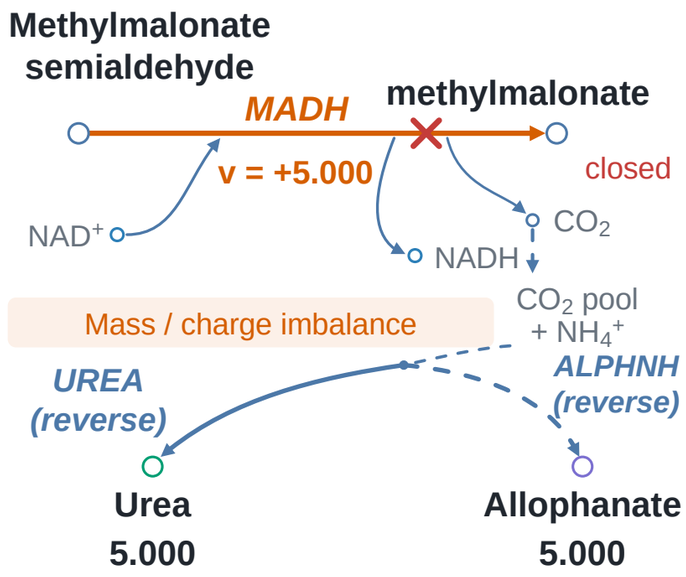
(3) Proton-supported ATP synthesis



Proton uptake supports NTP production

ATPS4r flux: 2.500

(4) Unsupported metabolite production

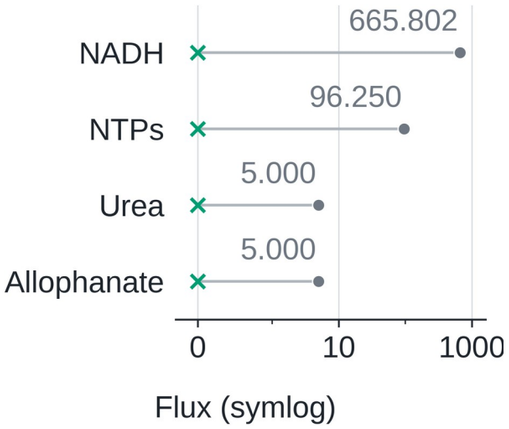


c

Post-MQC validation

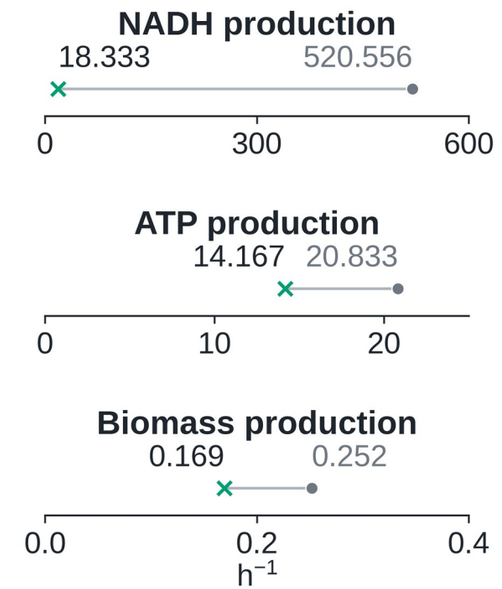
● Initial iCN718
× After MQC

No-carbon objectives



Matched-carbon re-test

Glucose uptake: 10
Other C sources closed



Active pFBA route · Problematic reaction · MQC stop · Metabolite · Membrane · Model-derived link

Flux: mmol gDCW^{-1} h^{-1}
c: cytoplasm; e: extracellular